Jatan Patel

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ABOUT

Final-year B.Tech in Computer Engineering (minor: Accounting & Finance). Applied-AI practitioner with a steady, respectful, and team-minded nature, building end-to-end solutions from data to deployment across vision, NLP, retrieval, and agent systems. Emphasizes careful experimentation, clear documentation, and useful tooling; currently advancing Generative AI and Reinforcement Learning with advanced statistics.

EDUCATION

Bachelor of Technology in Computer Engineering — **2022 – 2026**

*Nirma University (CPI: **8.05**)* — *Ahmedabad, India*

EXPERIENCE

Research Intern — **May 2025 – October 2025**

ISRO Project at Nirma University — *Ahmedabad, India*

- Designed a K-fold ensemble for lunar crater segmentation on Chandrayaan-2 OHRC tiles (segmentation-models-pytorch, UNet-family backbones); produced out-of-fold meta-features and trained a meta-learner that outperformed single models + HITL for checking and improvement.
- Boosted rim fidelity with a lightweight adversarial refiner (GAN + stability regularization) and ellipse-projection post-processing to reduce jagged edges and close broken rims.
- Built a full training/inference stack with Albumentations, AMP/GradScaler, and 4-fold CV; logged IoU/Dice/Accuracy, inference time, and model size across singles, pairs, triad (avg), and meta-learner.
- Delivered explainability utilities: per-model masks, pairwise blends, ensemble masks, plus "extra vs missed" overlays and heatmaps to visualize inter-model disagreement.
- Curated a georeferenced OHRC dataset from ISRO PRADAN; tiled imagery, created binary crater masks, and produced rasterized outputs for GIS; drafted a live human-in-the-loop loop for annotate → train → correct (Kaggle GPU + local RTX 3060).

PROJECTS

Live Weapon Detection System

- Real-time YOLOv8 detection (guns/knives) with accuracy of 96%, trained on a custom Roboflow dataset; hot-swappable `best.pt` for live inference.
- Flask backend with OpenCV MJPEG streaming; SQLite logging (type, confidence, camera, timestamp) and SMS alerts with cool-down.
- Responsive Bootstrap dashboard with live feed, camera selector, detection history; JS controls for start/stop and multi-camera switching.

Custom LLM Training & Alignment Pipeline

- Built an end-to-end GPT-style LLM from scratch (PyTorch): byte-level & BPE tokenizers, streaming dataset + next-token objective, custom training loop with AMP + gradient accumulation, warmup–cosine LR, checkpointing/resume, TensorBoard logs, and temperature/top-k/top-p sampling.
- Implemented modern Transformer optimizations: RMSNorm, RoPE, SwiGLU MLP, grouped-KV heads (GQA), KV cache with sliding-window + attention-sink for low-latency streaming generation; included small demos/visualizations to validate attention behavior.
- Delivered full alignment pipeline: Supervised fine-tuning utilities (prompt formatting, collators, length curriculum), a Transformer-based Reward Model for pairwise preferences, and RLHF trainers with PPO and GRPO (policy+value heads, advantages, KL control).
- Scaled & ensured correctness: Mixture-of-Experts (top-k routing with load-balancing aux loss), plus 20+ pytest unit tests covering attention math, causal masking, tokenizer/label-shift, LR scheduler, and RL losses; orchestration scripts to run tests, smoke-train, and sample.

Deep Learning Pipeline for Knotted Protein Generation and Toxicity Prediction (TopoToxiNet)

- Generate de-novo knotted protein sequences guided by known families and evaluate their biological safety through peptide/protein toxicity prediction.
- Curated datasets from PDB, UniProt, Pfam for sequence generation; fine-tuned ProtGPT2 for knotted sequence synthesis and integrated TopoToxiNet—an attention-regularized CNN–BiGRU multi-branch model—for toxicity assessment.
- Tech: Python, PyTorch/Transformers, Pandas, Matplotlib; sequence embeddings, KL regularization, ROC–PR evaluation metrics.
- End-to-end AI pipeline from data curation to generation and safety validation—producing novel knotted protein candidates with toxicity-aware screening for downstream structural modeling.

Buddy — Personal Voice Assistant for Windows (Python, OpenAI/Ollama, Vosk, PowerShell TTS)

- Wake-word assistant ("buddy") with one-shot and continuous modes; dual online/offline (OpenAI vs. Ollama+Vosk) with automatic privacy fallback.
- Speech stack: Google Speech + Vosk, ambient-noise calibration, short/long capture; async System.Speech TTS with barge-in and chunked playback; code-response suppression.
- Pluggable app launcher via `apps.json` (paths/args/UWP IDs/.lnk) plus discovery fallbacks (PATH, Start Menu, `shell:AppsFolder`, protocol handlers).
- Natural-language desktop control: open explicit paths (e.g., D:\ISRO), "open *folder* in *drive*", Explorer back/forward/up hotkeys, and quick site shortcuts.
- Ops & memory: timestamped .txt logs of all answers, .env config (wake word, models, mic, dirs, search depth), and admin voice commands ("reload/list apps", "add app path/uwp", "reset chat", "sleep/quit").

E-Grocery — MERN E-commerce Platform

- Full-stack MERN app; mobile-first React+Vite UI with Context API for predictable global state.
- Real-time cart updates, multi-address deliveries, server-side pagination/filtering for large catalogs.
- Seller Dashboard with inventory CRUD, price/stock controls, order tracking, and basic analytics (MongoDB aggregations).
- RESTful Express APIs on MongoDB with optimized schemas/indexes; deployed FE/BE on Vercel serverless for scalable, low-ops performance.

SKILLS

Programming: Python, C, C++, Java
Web: React.js, Django, Express.js, JavaScript
AI/ML/DL: PyTorch, TensorFlow, scikit-learn, OpenCV (detection, segmentation), Hugging Face, NLP (RNNs/LSTM/GRU/Transformers), Generative AI, Agentic AI, RL (foundations)
Data: Pandas, NumPy, SQL, Matplotlib, Tableau, Feature Engineering